

CSA0715-COMPUTER NETWORK

ASSESSMENT TOOL 1

SHORT ANSWER TEST:

1. A company needs to transfer a 150 MB design file from a client computer to a remote server over a network with a bandwidth of 50 Mbps. The Transport Layer divides the file into 1500-byte segments, while the Data Link Layer encapsulates each segment into frames by adding 40 bytes of header and trailer. Calculate the total number of segments and frames transmitted, the total transmission overhead introduced by the Data Link Layer, and estimate the total transmission time (ignoring propagation delay). Explain how the Transport and Data Link layers ensure reliable communication.

Solution:

Given:

- File size = 150 MB = $150 \times 10^6 = 150,000,000$ bytes
- Segment size = **1500 bytes**
- Header + Trailer = **40 bytes**
- Bandwidth = **50 Mbps**

Step 1: Number of Segments

$$\text{Number of Segments} = \frac{\text{File size}}{\text{Segment size}} = \frac{150,000,000}{1500} = 100,000$$

Step 2: Number of Frames

$$\text{Number of Frames} = \text{Number of Segments} = 100,000$$

Step 3: Total Overhead

$$\text{Total Overhead} = \text{Number of Frames} \times \text{Header + Trailer} = 100,000 \times 40 = 4,000,000 \text{ bytes} = 4 \text{ MB}$$

Step 4: Total Data

$$\text{Total Data} = \text{File size} + \text{Total Overhead} = 150 \text{ MB} + 4 \text{ MB} = 154 \text{ MB}$$

Step 5: Transmission Time

$$\text{Transmission Time} = \frac{\text{Total Data}}{\text{Bandwidth}} = \frac{154 \times 8}{50} = \frac{1232}{50} = 24.64 \text{ s}$$

Answer

- Segments = **100,000**
- Frames = **100,000**
- Overhead = **4 MB**
- Time = **24.64 s**

2. A Transport Layer protocol uses a sliding window size of 6 frames. Each frame carries 1024 bytes of user data. If the sender transmits 48 frames, calculate the number of transmission windows required. If one frame in every window is lost and must be retransmitted, determine the total number of retransmissions. Explain how flow control improves communication efficiency.

Solution:

Given

- Window size = 6
- Frames = 48

Step 1: Windows

$$= 48 \div 6 = \frac{48}{6} = 8$$

Step 2: Retransmissions

$$= 8 \times 1 = 8$$

Answer

- Windows = **8**
- Retransmissions = **8**

3. A 12,000-byte IP packet is transmitted through a router where the Maximum Transmission Unit (MTU) is 1500 bytes. Assuming each fragment requires a 20-byte IP header, calculate the total number of fragments created and the total header overhead introduced. Explain how the Network Layer ensures successful delivery of fragmented packets to the destination.

Solution:

Given

- Packet = 12,000 bytes
- MTU = 1500 bytes
- Header = 20 bytes

Step 1: Payload per Fragment

$$=1500-20=1480 \text{ bytes} = 1500-20 = 1480 \text{ bytes}$$

Step 2: Number of Fragments

$$= \frac{12000}{1480} = 8.1 = 9 \text{ fragments}$$

Step 3: Header Overhead

$$=9 \times 20 = 180 \text{ bytes}$$

Answer

- Fragments = **9**
- Header Overhead = **180 bytes**

4. A Transport Layer protocol uses a sliding window size of 6 frames. Each frame carries 1024 bytes of user data. If the sender transmits 48 frames, calculate the number of transmission windows required. If one frame in every window is lost and must be retransmitted, determine the total number of retransmissions. Explain how flow control improves communication efficiency.

Solution:

Step 1: Windows

$$= \frac{48}{6} = 8$$

Step 2: Retransmissions

$$= 8 \times 1 = 8$$

Answer

- Windows = **8**
- Retransmissions = **8**

5. A multimedia application transfers a 600 MB video file. The Session Layer inserts a synchronization checkpoint after every 60 MB of transmitted data.

During transmission, a failure occurs after 390 MB has been transferred. Determine the number of checkpoints created before failure and calculate the amount of data that must be retransmitted after recovery. Explain the importance of synchronization in the Session Layer.

Solution:

Given

- File = 600 MB
- Checkpoint = 60 MB
- Failure = 390 MB

Step 1: Checkpoints

$$= \frac{390}{60} = 6.5 \Rightarrow \text{Number of checkpoints} = 6$$

Step 2: Retransmitted Data

$$= 390 - 360 = 30 \text{ MB}$$

Answer

- Checkpoints = **6**
- Retransmission = **30 MB**

6. Before transmission, the Presentation Layer compresses a 900 MB file by 40%. The compressed data is encrypted, adding 5% overhead to the compressed size. Calculate the compressed file size, the encrypted file size, and the total percentage reduction compared to the original file. Explain how the Presentation Layer improves transmission efficiency and security.

Solution:

Given

- Original = 900 MB
- Compression = 40%
- Encryption = 5%

Step 1: Compressed Size

$$= 900 \times 60\% = 540 \text{ MB}$$

Step 2: Encrypted Size

$$=540 \times 1.05 = 567 \text{ MB} = 540 \times 1.05 = 567 \text{ MB}$$

Step 3: Reduction

$$= \frac{900 - 567}{900} \times 100 = 37\% = \frac{900 - 567}{900} \times 100 = 37\%$$

Answer

- Compressed = **540 MB**
- Encrypted = **567 MB**
- Reduction = **37%**

7. An FTP application transfers 3 GB of data over a network with an effective throughput of 120 Mbps. The Application Layer divides the data into requests of 3 MB each. Calculate the total number of requests generated and estimate the total transmission time in seconds. Discuss how the Application Layer provides reliable services to users.

Solution:

Given

- Data = 3 GB = 3000 MB
- Request = 3 MB
- Throughput = 120 Mbps

Step 1: Requests

$$= \frac{3000}{3} = 1000 = \frac{3000}{3} = 1000$$

Step 2: Transmission Time

$$= \frac{3000 \times 8}{120} = 200 \text{ s} = \frac{3000 \times 8}{120} = 200 \text{ s}$$

Answer

- Requests = **1000**
- Time = **200 s**

8. A packet experiences the following processing delays at different OSI layers: Application Layer = 4 ms, Transport Layer = 3 ms, Network Layer = 5 ms, Data Link Layer = 2 ms, and Physical Layer = 1 ms. The propagation delay is 18 ms,

and the transmission delay is 12 ms. Calculate the total end-to-end delay experienced by the packet and explain how each layer contributes to the communication process.

Solution:

Step 1: Processing Delay

$$=4+3+5+2+1=15 \text{ ms} = 4+3+5+2+1 = 15 \text{ ms}$$

Step 2: Total Delay

$$=15+18+12=45 \text{ ms} = 15+18+12 = 45 \text{ ms}$$

Answer

- Total Delay = **45 ms**

9. A network transfers 80 MB of useful data using frames of 1600 bytes, where each frame contains 80 bytes of protocol overhead added by different OSI layers. Calculate the total number of frames required, the total overhead transmitted, and the transmission efficiency as a percentage. Explain how protocol overhead affects network performance.

Solution:

Given

- Useful Data = 80 MB = 80,000,000 bytes
- Frame = 1600 bytes
- Overhead = 80 bytes

Step 1: Useful Data per Frame

$$=1600-80=1520 \text{ bytes} = 1600-80 = 1520 \text{ bytes}$$

Step 2: Frames

$$= \frac{80,000,000}{1520} = 52,631.57 \approx 52,632$$

Step 3: Overhead

$$=52,632 \times 80 = 4,210,560 \text{ bytes}$$

Step 4: Efficiency

$$= \frac{1520}{1600} \times 100 = 95\% = \frac{1520}{1600} \times 100 = 95\%$$

Answer

- Frames = **52,632**
- Overhead = **4,210,560 bytes**
- Efficiency = **95%**

10. A hospital transfers a patient's medical record of 240 MB through an IoT-enabled healthcare network. The Presentation Layer compresses the data by 30%, the Transport Layer divides the compressed data into 1400-byte segments, and the Data Link Layer adds 60 bytes of overhead to each frame. Calculate the compressed file size, the total number of segments transmitted, the total protocol overhead introduced, and the final amount of data transmitted across the Physical Layer. Explain how multiple OSI layers work together to ensure secure and reliable communication.

Solution:

Given

- Original = 240 MB
- Compression = 30%
- Segment = 1400 bytes
- Overhead = 60 bytes

Step 1: Compressed Size

$$= 240 \times 70\% = 168 \text{ MB} = 240 \times 70\% = 168 \text{ MB}$$

Step 2: Segments

$$= \frac{168,000,000}{1400} = 120,000 = \frac{168,000,000}{1400} = 120,000$$

Step 3: Overhead

$$= 120,000 \times 60 = 7,200,000 \text{ bytes} = 7.2 \text{ MB} = 120,000 \times 60 = 7,200,000 \text{ bytes} = 7.2 \text{ MB}$$

Step 4: Final Data

$$= 168 + 7.2 = 175.2 \text{ MB} = 168 + 7.2 = 175.2 \text{ MB}$$

Answer

- Compressed = **168 MB**
- Segments = **120,000**
- Overhead = **7.2 MB**

- Final Data = **175.2 MB**